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Climate Swings Drove Early Humans Out of Africa (and Back Again)

A new study details how climate change directed early modern humans' intricate dance among continents and pushes back their dispersal out of Africa to at least 100,000 years ago.

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THE TEXTBOOK NARRATIVE of human history tells us that between 70,000 and 60,000 years ago our earliest modern human ancestors traveled out of Africa on a journey that led them to nearby continents. But the factors that drove this mass exodus—as well as when it occurred and whether there was more than one big migration event—have long been points of spirited debate and contention. In a study published today in *Nature*, researchers report that dramatic climate fluctuations created favorable environmental conditions that triggered periodic waves of human migration out of Africa every 20,000 years or so, beginning just over 100,000 years ago.

These findings push back the clock on the earliest known arrivals of humans in Europe by tens of thousands of years and provide important clues about what drew early humans out of Africa and into Eurasia and beyond. The study further calls into question the deeply held dogma about a single, prominent migration out of Africa around 70,000 to 60,000 years ago. "What we cannot support is that there must have been a major single exit event from 70,000 to 60,000 years ago, which has become more or less the standard scenario for out of Africa," says Axel Timmermann, a climate scientist based at the University of Hawaii at Manoa and lead author of the study.

The new study suggests that there were "multiple opportunities for migrations of humans out of Africa, and that window of opportunity starts by at least 100,000 years ago," says Michael Petraglia, an archaeologist at the Max Planck Institute for the Science of Human History in Jena, Germany, who was not involved with the study. "Everything we're finding archaeologically makes sense relative to the climate simulations that [the new study] reports."

To understand the conditions that may have encouraged human dispersal, Timmermann and co-author Tobias Friedrich, a postdoctoral fellow also at the University of Hawaii at Manoa, developed one of the first climate simulation models that integrates paleoclimatic data across glacial and interglacial cycles, vegetation change, and human migration. By modeling climate variability over the last 125,000 years and accounting for sea-level changes and millennial-scale abrupt climate shifts, the team studied how

warm and wet periods in northern Africa led to lush vegetation and other conditions that were ripe for both mammals and hunter gatherers to move north and east.

"Our results reveal that human migration out of Africa and across the Sinai Peninsula and the Red Sea near Bab-el-Mandeb was not a single event as is often suggested but that it occurred in waves, and every 20,000 years or so Earth's axis wobble caused shifts in climate and vegetation in tropical and subtropical regions," asserts Timmermann. "Such shifts then opened up green corridors between Africa and the eastern Mediterranean and between Africa and the Arabian Peninsula, thus enabling *Homo sapiens* to leave northeastern Africa and to embark on their grand journey into Asia, Europe, Australia, and eventually to the Americas."

The study finds that humans traveled out of Africa in four waves across the Arabian Peninsula and the Levant region (the eastern Mediterranean). These waves occurred from 106,000 to 94,000 years ago, 89,000 to 73,000 years ago, 59,000 to 47,000 years ago, and 45,000 to 29,000 years ago—results that align well with a growing body of archaeological and fossil data. The wave that occurred approximately 50,000 years ago is likely the one that led to the population of the rest of the world. The new research also shows that *Homo sapiens* arrived simultaneously in southern China and Europe some 90,000 to 80,000 years ago.

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COLD, ARID CONDITIONS during the first half of the last ice age (110,000 to 60,000 years ago) were punctuated every 20,000 years by warm summers in the Northern Hemisphere. These climate shifts, triggered by the wobble of Earth's axis, created green corridors between Africa and Eurasia that set the stage for migratory waves of *Homo sapiens*. With the growth of lush grasses and shrubs, the expansion of animals and early humans out of Africa became possible. "If you imagine savannah biomes, they're perfect for herbivores such as cattle, oryx, and other four-hooved animals," explains Petraglia. "They migrate in those situations and eat off the grasslands, and then of course you have standing fresh water and aquatic resources in freshwater lakes."

In contrast, periods of low temperatures and extreme drought would have made human travel far less likely. Paleoclimatic models are in wide agreement that 60,000 to 70,000 years ago—an era that is often thought to be the main timeframe for modern human dispersal out of Africa—was one of the most extended drought periods in northern Africa, Saudi Arabia, and the eastern Mediterranean in the last 125,000 years, says Timmermann. “Walking into the Arabian Peninsula around 60,000 to 70,000 years ago,” he remarks, “would have been a bad choice.”



The oryx is one of the many mammal species that may have led early modern humans out of—and in some cases back into—Africa during migrations over the last 100,000 years.
Benjamin Hollis/Flickr

Yet, Timmermann notes, just 10,000 years earlier (roughly 80,000 years ago) would have made for "wonderful conditions" for migration into Saudi Arabia and the eastern Mediterranean region.

The team's model suggests a more fluid exchange between Africa and nearby continents rather than a one-way exodus out of Africa. "Chasing prey through the green savannah in northeastern Africa, Sinai, the eastern Mediterranean, and the Arabian Peninsula, early *Homo sapiens* would not even have recognized the difference between Africa and Eurasia," Timmermann says. "These migration corridors work in two directions."

Peter deMenocal, a paleoclimatologist based at Columbia University, who was not involved with the study, gave the analogy of climate acting like a pump and a valve at different times. Long periods of cold, dry, inhospitable conditions closed the valve on migration. But when warm, moist, tropical conditions set in, they opened the valve, connecting adjacent regions that were previously out of reach. "I think what's great about this paper is that it really clarifies the role of orbital climate change in pacing the peopling of the world," deMenocal adds. Once conditions were right for human migration, "they got out of Dodge."

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ONE OF THE more surprising findings of the new study is that it shows wisps of human migration into southern Europe at around 80,000 to 90,000 years ago—approximately 45,000 years earlier than the oldest fossil evidence of early humans in the region. "This [earlier migration] is a plausible scenario, but it's so much at odds with what we know about the fossil record in Europe, so that came as a big surprise to me," says Timmermann. The time discrepancy, he adds, "still needs to be reconciled."

Three genomics studies also published today in *Nature* add insight and complexity to the timing of early human dispersal. The new papers both converge and diverge with the paleoclimate study in important ways, observes Petraglia, who was one of a suite of authors on a study that analyzed a dataset of nearly 500 genomes from 148 populations worldwide. The study Petraglia contributed to provides additional evidence that modern humans emerged from Africa approximately 100,000 years ago. “We have a convergence or a correspondence between the theoretical arguments of these climate simulations and our own on-the-ground evidence. So there’s a perfect marriage between the two very independent studies.”

Although the new paleoclimate study opens the door to the possibility of earlier and more frequent human migrations, it’s unlikely that those dispersals played a big role in populating the world, says Petraglia. Research suggests that the majority of the earliest human groups eventually faded away after they arrived in Eurasia and that most people alive today can trace their ancestral lineage to the migration that occurred between 59,000 and 47,000 years ago. “Actually, there were a lot of experiments in the past, in a sense. Some populations might have been more successful than others. We never think of lineages of *Homo sapiens* potentially going extinct, but that’s very possible.”



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Prior to becoming the managing editor of SAPIENS in May 2015, Amanda Mascarelli spent more than a decade as a freelance science journalist. She has written about oil spills, autism, the neuroscience of magic, the biological complexity of sex and gender, and many other topics, and her reporting has taken her around the world. Her work appears in *Audubon*, *Nature*, *Science*, *Science News for Students*, *The Los Angeles Times*, *The New York Times*, *The Washington Post*, and elsewhere. Mascarelli earned an undergraduate degree in biology and a master's in journalism, and spent several years doing biology research before becoming a writer and editor. She lives with her husband and three children in Denver, Colorado.